

ILHA FERNANDO DE NORONHA, Brazil

WMO: 824000

Lat: 3.855S

Lon: 32.423W

Elev: 59

StdP: 100.62

Time Zone: -2.00 (W02)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.9	23.3	19.8	14.7	25.5	20.1	14.9	25.5	8.4	26.1	7.7	25.9	3.1	150	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.2	31.2	25.8	30.9	25.8	30.5	25.6	26.7	30.0	26.3	29.7	26.1	29.6	4.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	21.2	29.2	25.2	20.5	28.6	25.1	20.4	28.5	84.0	29.9	82.5	30.1	81.7	29.7	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.5	6.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			18.5	28.8	1.5	1.1	17.4	29.6	16.6	30.3	15.8	31.0	14.7	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	27.7	27.6	27.6	27.1	27.5	26.9	26.4	26.3	26.7	27.1	27.5	27.8
	DBStd	0.86	0.73	0.85	0.87	1.00	0.67	0.74	0.69	0.64	0.51	0.56	0.60	0.47
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6271	550	494	544	514	542	508	509	505	500	529	524	551
	CDD18.3	3229	292	261	286	264	283	258	251	247	250	271	274	293
	CDH23.3	29822	2920	2633	2854	2455	2784	2347	2028	1916	2058	2387	2569	2870
	CDH26.7	7191	800	709	762	580	692	490	361	327	437	582	678	773
Wind	WSAvg	4.0	3.9	3.5	3.3	3.0	3.4	3.8	4.1	4.5	4.6	4.6	4.7	4.5
Precipitation	PrecAvg	999	62	104	170	248	176	98	63	23	10	6	8	30
	PrecMax	1506	193	287	330	471	433	404	184	60	28	19	39	155
	PrecMin	393	6	11	43	121	23	12	8	7	2	2	1	2
	PrecStd	299	52	62	75	95	99	81	41	14	5	4	9	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.8	31.2	31.2	31.1	30.2	30.0	29.6	30.2	30.8	31.1	31.2
		MCWB	25.6	25.6	25.9	26.4	26.3	25.8	24.7	24.2	24.4	24.9	25.0	25.4
	2%	DB	31.0	31.0	31.0	30.6	30.2	30.0	29.2	29.1	29.9	30.2	30.4	30.8
		MCWB	25.6	25.7	26.0	26.2	26.0	25.7	24.5	24.2	24.3	24.7	25.0	25.6
	5%	DB	30.5	30.5	30.4	30.1	30.1	29.5	29.0	28.9	29.2	29.9	30.1	30.2
		MCWB	25.4	25.6	25.9	26.1	25.9	25.3	24.5	24.2	24.2	24.6	24.8	25.2
	10%	DB	30.1	30.0	30.0	29.8	29.8	29.0	28.7	28.3	29.0	29.2	29.9	30.0
		MCWB	25.3	25.6	25.9	26.0	25.8	25.0	24.3	24.0	24.1	24.2	24.7	25.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	26.7	27.1	27.1	27.0	26.2	25.7	25.4	25.6	26.1	26.0	26.4
		MCDB	30.4	30.3	30.3	30.3	30.2	29.4	28.2	28.5	29.1	29.6	30.5	30.3
	2%	WB	26.0	26.2	26.5	26.6	26.4	25.9	25.2	24.9	25.0	25.3	25.5	26.0
		MCDB	30.1	29.9	29.9	29.7	29.7	29.2	28.2	28.3	28.8	29.3	29.9	30.0
	5%	WB	25.7	26.0	26.2	26.2	26.1	25.6	25.0	24.6	24.6	24.9	25.1	25.6
		MCDB	29.6	29.7	29.5	29.3	29.4	28.8	28.1	27.9	28.5	29.0	29.5	29.5
	10%	WB	25.4	25.7	26.0	26.0	25.9	25.3	24.6	24.3	24.3	24.5	24.8	25.2
		MCDB	29.2	29.1	29.3	29.1	29.0	28.4	27.6	27.6	28.1	28.5	29.1	28.9
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.2	5.1	5.0	4.9	4.7	4.7	4.7	5.1	5.3	5.4	5.1
		MCDBR	5.5	5.6	5.6	5.5	5.2	5.1	5.0	5.0	5.3	5.6	5.5	5.2
		MCWBR	2.4	2.5	2.7	3.0	2.7	2.6	2.5	2.6	2.6	2.6	2.4	2.4
	5% WB	MCDBR	5.2	5.2	5.2	5.3	5.0	4.9	4.7	4.7	5.1	5.3	5.5	5.1
MCWBR		2.5	2.6	2.7	3.0	2.7	2.7	2.6	2.7	2.7	2.7	2.5	2.4	
Clear-Sky Solar Irradiance	taub	0.410	0.417	0.412	0.401	0.401	0.398	0.400	0.413	0.437	0.408	0.400	0.414	
	taud	2.508	2.506	2.537	2.596	2.576	2.565	2.551	2.481	2.377	2.472	2.516	2.478	
	Ebn at Noon	926	919	912	899	873	862	866	875	876	916	929	918	
	Edh at Noon	113	114	109	99	97	96	99	110	126	116	111	115	
All-Sky Solar Radiation	RadAvg	6.37	6.19	6.05	5.45	5.45	5.38	5.55	6.19	6.62	6.92	6.94	6.62	
	RadStd	0.44	0.47	0.42	0.40	0.34	0.32	0.31	0.16	0.18	0.12	0.12	0.26	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page